

Case Study

HR Employee Attrition Analysis

Problem Statement

The HR leadership team has observed increasing employee attrition across the organization, but the reasons behind employees leaving remain unclear. Some departments are losing talent faster than others, and management doesn't know whether compensation, job satisfaction, work-life balance, commute distance, or career growth is the primary driver.

Without identifying the root causes, retention strategies become reactive instead of data-driven, leading to higher recruitment costs, productivity loss, and declining employee morale.

The objective of this analysis is to identify the key factors influencing employee attrition and provide actionable recommendations to improve employee retention.

Data Overview

- **Source:** IBM HR Employee Attrition Dataset
- **Records:** ~1,470 employees
- **Tools Used:** Excel, SQL, Power BI

The dataset contains employee demographics, salary, department, job role, overtime, satisfaction scores, commute distance, years at the company, and attrition status etc.

Business Questions

1. What's the overall attrition rate, and how does it vary by department?
2. Does attrition correlate with low job satisfaction scores?
3. Is attrition higher for employees with longer commute distances?
4. How does attrition differ by salary band?
5. Is there a tenure "danger zone" (e.g., year 1-2) where most people leave?
6. Which job role has the highest attrition?

Tool Stack

Excel (Data Cleaning) → **SQL** (Business Analysis) → **Power BI** (Interactive Dashboard)

Data Cleaning (Excel)

Performed data validation and quality checks before analysis.

Cleaning Steps

- Verified there were no duplicate Employee IDs.
- Checked for missing or blank values.
- Corrected inconsistent text formatting.
- Verified Attrition values were limited to "Yes" and "No".
- Created an **Attrition Flag** column for easier aggregation.
- Created **tenure bands**:
 - 0–2 Years
 - 3–5 Years
 - 6–10 Years
 - 10+ Years
- Created Monthly Income Bands.

Analysis (SQL)

Key Findings

1. What's the overall attrition rate, and how does it vary by department?

Overall employee attrition rate is **16.12%**, indicating that approximately **1 in 6 employees** leave the organization.

Total_Employees	Attrited	Attrition_Rate_Pct
1470	237	16.12

Department-wise Attrition

Department	Attrited	Attrition_Rate_Pct
Sales	92	20.63
Human Resources	12	19.05
Research & Development	133	13.84

The **Sales department** has the highest attrition rate at **20.63%**, followed by **Human Resources (19.05%)** and **Research & Development (13.84%)**. While R&D experienced the greatest number of employee departures (133), the proportion of employees leaving is highest in Sales, indicating a greater retention challenge.

2. Does attrition correlate with low job satisfaction scores?

Employees with a Job Satisfaction score of **1** have an attrition rate of **22.84%**, compared to only **11.33%** among employees with the highest satisfaction score.

Employee satisfaction is one of the strongest predictors of attrition.

JobSatisfaction	Satisfaction_Label	Total_Employees	Attrited	Attrition_Rate_Pct
1	Low	289	66	22.84
3	Very High	442	73	16.52
2	Medium	280	46	16.43
4	Very High	459	52	11.33

3. Is attrition higher for employees with longer commute distances?

Employees commuting more than **16 km** show an attrition rate of **20.67%**, approximately **7 percentage points** higher than employees living closer to the workplace.

Longer commuting distances contribute to increased employee turnover.

Commute_Band	Total_Employees	Attrited	Attrition_Rate_Pct
Far (16+)	329	68	20.67
Mid (6-15)	509	82	16.11
Near (1-6)	632	87	13.77

4. How does attrition differ by salary band?

Employees earning less than **₹3,000/month** experience an attrition rate of **26.61%**, nearly **2×** higher than employees in the highest income band.

Lower compensation appears strongly associated with higher turnover.

Salary_Band	Total_Employees	Attrited	Attrition_Rate_Pct
1. Low (<3k)	395	113	28.61
2. Mid (3k-7k)	640	77	12.03
3. High (7k-12k)	240	36	15.00
4. Very High (12k+)	195	11	5.64

5. Is there a tenure "danger zone" (e.g., year 1-2) where most people leave?

Employees with **0–2 years** of service have the highest attrition at **29.82%**, making early tenure the highest-risk period.

Retention efforts should focus on improving the employee experience during the first two years.

Tenure_Band	Total_Employees	Attrited	Attrition_Rate_Pct
0-2 Years	342	102	29.82
3-5 Years	434	60	13.82
5-10 Years	448	55	12.28
10+ Years	246	20	8.13

6. Which job role has the highest attrition?

The **Sales Representative** position records the highest attrition rate at **39.76%**, despite an average monthly salary of **₹2626.00**.

This indicates that factors beyond salary—such as workload, career growth, or job expectations—may be driving employee departures.

JobRole	Total_Employees	Avg_Monthly_Income	Attrited	Attrition_Rate_Pct
Sales Representative	83	2626.00	33	39.76
Laboratory Technician	259	3237.17	62	23.94
Human Resources	52	4235.75	12	23.08
Sales Executive	326	6924.28	57	17.48
Research Scientist	292	3239.97	47	16.10
Manufacturing Director	145	7295.14	10	6.90
Healthcare Representative	131	7528.76	9	6.87
Manager	102	17181.68	5	4.90
Research Director	80	16033.55	2	2.50

7. Other factors Analysis

-Analysis by Gender And Martial status

Single – Male & Female records the highest attrition rate at **29.94% and 23.62%**, compared to Married Male & Female.

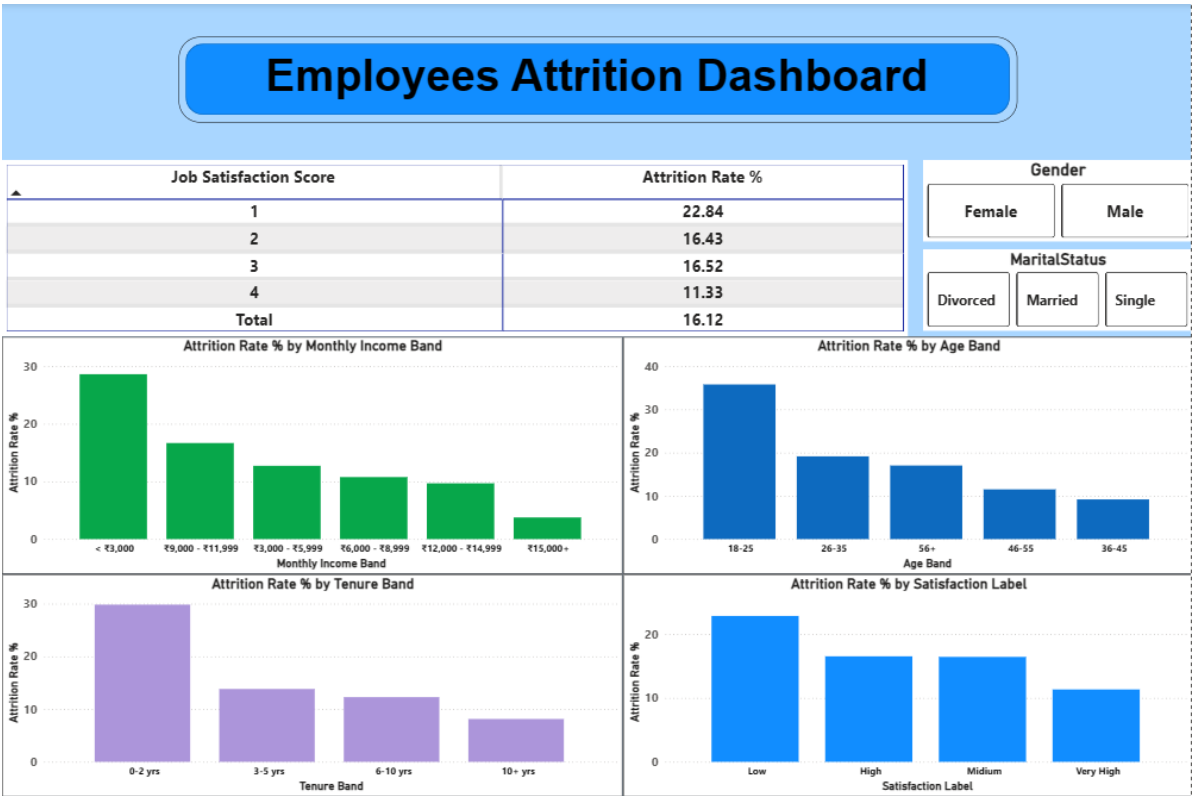
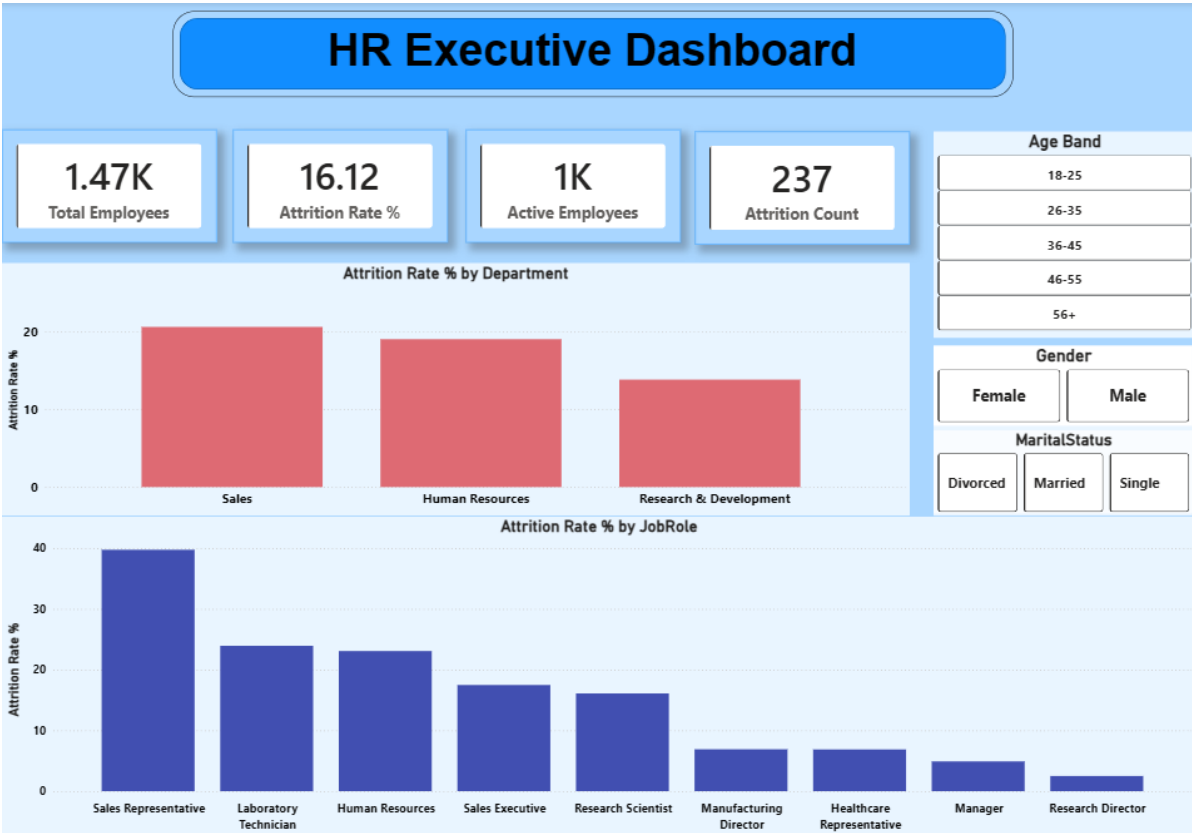
Gender	MaritalStatus	Total_Employees	Attrited	Attrition_Rate_Pct
Male	Single	271	73	26.94
Female	Single	199	47	23.62
Male	Married	401	53	13.22
Male	Divorced	210	24	11.43
Female	Married	272	31	11.40
Female	Divorced	117	9	7.69

- Analysis By Age Group

The **Early-Career Group (18-25)** Shows higher attrition rate of **35.77%**, compare to any other age group.

Age_Band	Total_Employees	Attrited	Attrition_Rate_Pct
18–25 Early-career employees	123	44	35.77
26–35 Young professionals (typically highest att...	606	116	19.14
56+ Pre-retirement employees	47	8	17.02
46–55 Senior professionals	226	26	11.50
36–45 Mid-career employees	468	43	9.19

Dashboard (Power BI)



Recommendations

1. Strengthen Early Employee Retention

Employees within their first two years have the highest attrition. Introduce a structured onboarding program, mentoring, and regular manager check-ins during the first six months.

2. Review Compensation Strategy

Conduct salary benchmarking for employees in the lowest income band to ensure compensation remains competitive and aligned with market standards.

3. Improve Employee Satisfaction

Launch quarterly employee pulse surveys and implement action plans for departments reporting consistently low satisfaction scores.

4. Reduce Commute Burden

Offer hybrid work options, flexible working hours, or transportation assistance for employees with longer commuting distances.

5. Manage Overtime More Effectively

Monitor excessive overtime, redistribute workloads where necessary, and encourage healthier work-life balance to reduce burnout.

6. Focus on High-Risk Job Roles

Conduct structured exit interviews for job roles with the highest attrition to identify recurring issues related to career progression, management, or compensation.

7. Build a Predictive Retention Strategy

Use employee satisfaction, tenure, overtime, and compensation data to identify employees at high risk of leaving and proactively intervene before resignation.